

### Problem 10.23

You are given that  $\delta = 0.05$ . Calculate the amount that must be invested at the end of 10 years to have an accumulated value at the end of 30 years of \$1,000.

### Solution

Let  $X$  be the amount invested at the end of year 10. It accumulates from year 10 to year 30, a period of

$$30 - 10 = 20 \text{ years.}$$

With a constant force of interest  $\delta = 0.05$ , the accumulation factor over 20 years is

$$e^{\delta(20)} = e^{0.05(20)}.$$

Therefore,

$$Xe^{0.05(20)} = 1000.$$

Solving for  $X$ ,

$$\begin{aligned} X &= 1000e^{-0.05(20)} \\ &= 1000e^{-1} \\ &\approx 367.8794412. \end{aligned}$$

Thus, the required investment at the end of year 10 is approximately

$$\boxed{\$367.88}.$$